

Do LLM Attribution Metrics Transfer?

Auditing Retrieval-Augmented Generation Evaluation Across Datasets and Constructs

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Abstract

Practice often treats automatic metrics for attribution in LLM retrieval-augmented generation as interchangeable. We audit eight automatic scorers — lexical, embedding, and BERTScore baselines alongside entailment/grounding-trained models (clean and FEVER NLI, the checker MiniCheck) — across three evaluation constructs (provenance/topicality, generated-answer attribution, and fact-check entailment), asking whether any scorer *transfers*: stays within the 95% confidence interval of the best audited scorer on *every* dataset of a multi-dataset construct. In the construct with the most multi-dataset **human**-labeled coverage — generated-answer attribution (AttributionBench’s four source datasets, $n=1610$, with independent HAGRID, $n=2150$) — none of the audited automatic scorers does: the per-dataset metric rankings invert (Kendall $\tau = -0.64$, $p=0.031$ on AttributedQA vs. LFQA), and an off-the-shelf NLI scorer that is best on short-claim AttributedQA (AUROC 0.90) collapses to AUROC 0.53 (chance) on long-form LFQA, where BERTScore wins (0.91); the reversal persists under the tested truncation settings. This instability has a concrete decision cost: a naive “best-on-average” rule for choosing an evaluator *fails* leave-one-dataset-out (mean held-out regret 0.172 AUROC, worse than fixing one scorer), so metric choice should be validated on the target dataset rather than assumed from performance elsewhere. A prompt-based LLM judge avoids the chance-level collapses the automatic scorers suffer (no LFQA collapse) but is not uniformly best, $\sim 100\times$ costlier, and non-deterministic — relocating, not removing, the validation burden.

1 Introduction

Retrieval-augmented language models are increasingly evaluated for *grounding*: whether each claim in an answer is supported by the provided evidence. A growing toolbox of automatic scores

— lexical overlap, sentence-embedding similarity, NLI/entailment — stands in for human grounding judgments, and practitioners often report them interchangeably, as if “the attribution metric went up” meant the same thing regardless of metric or benchmark. By “LLM attribution metrics” we mean metrics evaluating attribution in LLM/RAG outputs (lexical, embedding, NLI/checker, and prompted LLM-judge variants); our core audit is the cheaper *automatic* scorers, with the prompted LLM judge as a boundary case. We show this assumption is unsafe at two levels. Coarsely, metric-family performance varies across the audited *constructs* and datasets. More sharply, *within* a single construct the best metric is not stable across its datasets, so no audited automatic scorer passes our core generated-answer attribution portability screen. This is a claim about *evaluator selection* at the item level (which scorer to trust on a given dataset), not about the ranking of complete RAG systems, which we do not study (see §6).

Concretely, we audit eight metrics — off-the-shelf scores with no attribution-specific fine-tuning (lexical Jaccard, MiniLM and MP-Net cosine, their fixed blend, BERTScore) and trained entailment/grounding models (clean-MNLI NLI, FEVER NLI, and the purpose-built checker MiniCheck (Tang et al., 2024); a lexical/semantic/combined/NLI subset on the provenance ranking task) — across three evaluation constructs the literature routinely conflates: (i) **provenance/topicality** — does a score rank a relevant passage top (ASQA (Stelmakh et al., 2022), MS MARCO (Nguyen et al., 2016), HotpotQA (Yang et al., 2018); passage-ranking labels, not sentence/claim-level support); (ii) **generated-answer attribution** — does a score predict *human* judgments of whether a generated answer sentence is supported by its cited evidence (AttributionBench (Li et al., 2024)); (iii) **fact-check entailment** — does a score predict *human* support

labels on short edited claims (VitaminC (Schuster et al., 2021)). At the coarse, cross-construct level metric-family performance differs on the audited benchmarks (Table 2): the best off-the-shelf relevance score per provenance gate reaches 0.73–0.97 top-1 accuracy (within overlapping bootstrap CIs of NLI on MS MARCO/HotpotQA, though NLI is unreliable on ASQA at 0.585), whereas entailment-trained models score above all no-fine-tuning baselines on fact-check claims (VitaminC, 0.81–0.89 vs. surface $\approx$ 0.60 AUROC); this coarse contrast is bounded to the audited benchmarks and partly confounded with domain shift (§6). The sharper finding is *within* generated-answer attribution: across AttributionBench’s four constituent datasets the best metric is not stable (the per-dataset rankings invert, Kendall $\tau = -0.64$, $p=0.031$ on AttributedQA vs. LFQA). A naive NLI scorer is best on AttributedQA (AUROC 0.90) yet drops to 0.53 (chance = 0.50) on LFQA, where BERTScore reaches 0.91 — though on a *different* long-form set (HAGRID) the same NLI scorer is best (0.80), so the failure is dataset-specific and not explained by long-form *answers* alone (long *evidence* remains a candidate; §4).

This matters because practitioners silently read success on one construct, or on one benchmark, as evidence of grounding in general. Prior benchmarking work establishes that automatic attribution metrics disagree with humans and that no single metric wins everywhere (Honovich et al., 2022; Dziri et al., 2022; Yue et al., 2023; Li et al., 2024); beyond that qualitative “they disagree,” our contribution is decision-relevant: an operational *portability* criterion, a leave-one-dataset-out *regret* that prices committing to one default, and a naive cross-dataset selector that *fails* (0.172 regret) — a validation discipline, on a controlled audit with human labels on the in-domain attribution and fact-check sides and a clean separation from proxy-labeled provenance (mechanism discovery is out of scope). This is an evaluation-resource/cautionary paper, not a new-metric paper. Our central contribution is the cross-dataset evaluator-selection audit; the rest are supporting diagnostics and boundary probes:

- A **cross-dataset audit** (Section 4, Table 2) showing metric-family differences across the audited constructs and datasets, but that *within* in-domain answer attribution no single *automatic* scorer is consistently near-best across datasets (rankings discordant, Kendall $W=0.07$, $p=0.029$): a raw

MNLI scorer is best on AttributedQA (0.90) yet AUROC 0.53 (chance = 0.50) on the LFQA dataset — though best again on the independent long-form HAGRID set (0.80) — while a purpose-built checker (MiniCheck) has the lowest mean leave-one-dataset-out regret among the eight automatic scorers *tested*, yet still leaves a measurable per-dataset regret (mean 0.044 AUROC, CI [0.014, 0.072]).

- As supporting diagnostics, a training-free **sentence-unit provenance/topicality diagnostic** and a **confusable-evidence stress-test protocol** (Sections 3, 3.1): under confusable evidence (high-overlap distractors from other examples), the semantic component separates co-provenant passages from distractors where lexical overlap is fooled. We benchmark an off-the-shelf NLI entailment head against it and find NLI benchmark-dependent, not uniformly best — a construct-(i) diagnostic, not a general-purpose grounding metric.
- On **validation discipline** (Section 4): a naive “best-on-average” selection rule fails leave-one-dataset-out (mean regret 0.172, worse than the lowest-mean-regret fixed metric among those tested), so metric choice must be validated on the target dataset; and a **cross-family LLM-annotation** probe (Opus 4.8 + GPT-5.4, $n=160$) showing high inter-model but only moderate, dataset-dependent human agreement — usable as a corroborative proxy only in the easiest (short-factoid) dataset, bounding off-the-shelf LLM label substitution.

An auxiliary rewrite-sensitivity probe appears in App. C; it does not establish improved grounding.

In short, “attribution metric” names a family of signals capturing different constructs (provenance, topical relatedness, human-judged support); metric suitability varies across the audited settings, and *within* answer attribution no audited automatic scorer passes our portability screen across the tested datasets. Evaluations should report the dataset and construct they target and prefer a checker validated on that dataset, not treat any single metric as a universal proxy for grounding.

2 Related Work

Correctness vs. faithfulness. An answer can be correct while its citations do not support its claims

(Rashkin et al., 2023; Liu et al., 2023): a faithfulness gap distinct from factual correctness (Ji et al., 2023), acute for long-form answers exact match cannot score (Fan et al., 2019) and worsened by uneven use of mid-context evidence (Liu et al., 2024).

Attribution and citation quality. AIS (Rashkin et al., 2023) formalizes source support, and audits find fluent answers often carry unsupported citations (Liu et al., 2023). ALCE (Gao et al., 2023) introduced NLI-based citation recall/precision (with QAMPARI (Amouyal et al., 2023)); FActScore (Min et al., 2023) decomposes into atomic facts; Attribute-First (Slobodkin et al., 2024) and LongCite (Zhang et al., 2024) produce attributable text; AttrScore (Yue et al., 2023) and RAGAS (Es et al., 2024) judge support with LLM/entailment models. Attribution-Bench (Li et al., 2024) benchmarks such evaluators against human labels — but, crucially, only *trained/LLM* evaluators, not the off-the-shelf lexical/embedding baselines we find within overlapping CIs in-domain. These methods need entailment/LLM judges or modify generation; our scorers include a deterministic sentence-unit one (Reimers and Gurevych, 2019), and our contribution is the cross-dataset *audit*.

RL for grounding. Prior work trains models to cite evidence via rewards, from verified-quote RLHF (Menick et al., 2022) to citation rewards (Huang et al., 2024), gated sufficiency/abstention rewards (Zhao et al., 2026), and verifiable-reward training (Sim et al., 2025). These *fix the model*; we *diagnose the evaluation*, and our null result (App. C) is consistent with their report that answer-level metrics are hard to move.

Distractors and robustness. Contextual distractors cause large drops (Lee et al., 2026), especially *highly semantically related* ones (Wu et al., 2024), and grounding can fail even in instruction-tuned models under adversarial conditions (Koneru et al., 2026). Our confusable-evidence test (Section 3.1) is a controlled, passage-ranking instance.

3 A Sentence-Unit Provenance-Ranking Score

We score how well each sentence of an answer is *matched* to a set of candidate passages, for the provenance-ranking diagnostic of construct (i) (distinct from the human-support task of constructs ii, iii). Let an answer be split into sentence

units $u_1, \dots, u_m$ and let the evidence be passages $p_1, \dots, p_k$. For a unit u and passage p we compute a combined relevance score

$$s(u, p) = 0.40 \text{lex}(u, p) + 0.45 \text{sem}(u, p) + 0.10 h(u) + 0.05 c(u, p), \quad (1)$$

where lex is token Jaccard, sem is sentence-transformer cosine (Reimers and Gurevych, 2019) (all-MiniLM-L6-v2), the *answer-hint* $h(u)$ is the fraction of gold short answers in the unit, and the *citation bonus* $c(u, p)=1$ when u cites p . A unit’s score is $\max_p s(u, p)$. It is **training-free** (only a small frozen encoder).

The score is a fixed *provenance* diagnostic for construct (i), **not** a proposed general-purpose grounding metric (far from best on the human-support tasks of Section 4) and **not** the paper’s central contribution: the human-labeled cross-dataset audit of Section 4. We report the single canonical weighting, not a tuned one. The answer-hint $h(u)$ is partly label-aware, but it does not affect the confusable ranking: $h(u)$ is constant within a unit (so it cannot reorder that unit’s candidate passages), and the citation term $c(u, p)=0$ throughout because ASQA gold answers carry no inline citations. The confusable test thus ranks on lexical/semantic only and the central result needs no gold-label access (ablated below).

3.1 Provenance/topicality (construct i)

The first construct asks whether a score routes an answer sentence to the passage it came from. On ASQA we form a gold-reference vs. unadapted-model contrast: the gold long-form answer vs. an unadapted Qwen2.5-3B-Instruct generation over the example’s ≥ 2 passages (623 dev examples; the confusable test uses a stricter ≥ 3 -passage filter, 339).¹ Holding short-answer correctness roughly equal (98 examples within 0.15 recall), the combined provenance gap is +0.160, CI [+0.133, +0.189] (full-dev +0.164), and is not a hint artifact (zeroing the hint weight moves the matched gap only to +0.153, CI excluding 0; App. B). So provenance-sensitive scores move even when correctness does not — but this is provenance sensitivity, not human-judged support.

Confusable-evidence stress test. To test the “just lexical overlap” objection, for each gold answer sentence we build a pool from the unit’s own

¹Two small ASQA checks (App. A) are *internal consistency* only (circular labels), so we do not rely on them.

Scorer	Top-1 prov. acc.	Fooled rate
Lexical-only	0.775	0.225
Semantic-only	0.934	0.066
Combined (canon.)	0.919	0.081
NLI entailment [†]	0.585	0.415

Table 1: Confusable-evidence stress test on ASQA (top-1 averaged across four seeds; best in bold; semantic encoder all-mpnet-base-v2, $k=5$ distractors). Lexical / semantic / combined are on the full usable dev pool (339 ex / 1309 units). The combined–lexical gap is $+0.144$, 95% CI $[+0.122, +0.165]$: all four independent-seed gaps ($+0.144, +0.144, +0.144, +0.143$) have CIs above 0. [†]The NLI row is computed on the 200-example-per-seed subsample (3 seeds, ≈ 760 units each) for cost, so its absolute value is not directly pooled with the others; the lexical and semantic values on that same subsample are 0.786 and 0.938, so NLI’s 0.585 is well below both on matched data. Ground truth is a passage-provenance proxy; distractors are highest-lexical-overlap passages from other examples. NLI = an off-the-shelf DeBERTa-v3 MNLI/FEVER/ANLI entailment head (passage→sentence) on this ranking task, not full ALCE citation evaluation.

ASQA passages plus $k=5$ *confusable distractors* (highest lexical-overlap passages from *other* examples). The label is a **provenance proxy** (did the top passage come from the unit’s own example), not a human groundedness judgment. We report top-1 provenance accuracy and the *fooled rate* (a confusable distractor outranks every co-provenant passage); ranking uses lexical/semantic only.

The semantic component survives; lexical is fooled; NLI does not help. On the full dev pool (339 examples, 1309 units; semantic encoder all-mpnet-base-v2 for this confusable test), top-1 provenance accuracy is 0.775 lexical-only, 0.934 semantic-only, 0.919 combined (combined–lexical $+0.144$, CI $[+0.122, +0.165]$, all four fresh-example seed gaps > 0 : 0.144, 0.144, 0.144, 0.143); the fooled rate drops from 22.5% (lexical) to 8.1% (combined) and 6.6% (semantic). An off-the-shelf DeBERTa MNLI/FEVER/ANLI scorer reaches only 0.585 here — below lexical, far below semantic — a construct mismatch (ASQA sentences *synthesize* across passages; cf. Laban et al., 2022), not a verdict on entailment in general. Sweeping the lexical weight α (Fig. 1, App. B) confirms the canonical blend ($\alpha \approx 0.47$) is past the knee, so we do **not** claim it optimal; this is specific to construct (i), where semantic carries the signal —

on human-judged support (ii, iii) it is *not* dominant.

4 The Cross-Dataset Audit

We make our central point in two steps: coarsely, metric-family performance differs across the audited constructs and datasets; more sharply, *within* a construct the best metric stays unstable across datasets. For this audit we use *portable* (its ranking *transfers*) as one near-best screen, operational *for metric selection*: a metric stays within the 95% CI of the best *audited* metric on *every* dataset of a construct. We do not offer this as a universal definition of transfer, and our consequence is at the level of choosing an evaluator, not ranking generators (system-level transfer is future work; see Limitations). In our data the same non-portability reading is corroborated by discordant per-dataset metric rankings (concordance $W=0.07$, $p=0.029$; the AttributedQA-vs-LFQA ranking inverts, Kendall $\tau = -0.64$, $p=0.031$), a non-zero leave-one-dataset-out regret (mean 0.044 AUROC for the best fixed choice), paired-bootstrap sign flips excluding zero (NLI–BERTScore = -0.378 on LFQA vs. $+0.213$ on AttributedQA), the ϵ -regret sweep (no audited metric transfers at $\epsilon \leq 0.05$; only MiniCheck at $\epsilon=0.10$), and the further sensitivity analyses in App. F. Dataset-dependent winners also occur in an independent replication excluding the overlapping ExpertQA and LFQA datasets ($n=7336$, App. F). We audit eight metrics — off-the-shelf scores with no attribution-specific fine-tuning (lexical Jacard, MiniLM and MPNet cosine, their blend, BERTScore (Zhang et al., 2020)) and trained models (clean non-FEVER MNLI NLI, FEVER NLI, and the purpose-built checker MiniCheck (Tang et al., 2024)) — on the two support constructs, and a lexical/semantic/combined/NLI subset on the provenance ranking task; we never pool the three constructs: **(i) provenance/topicality** (ASQA, MS MARCO, HotpotQA: rank candidate passages and check the top one; ASQA uses a provenance proxy, MS MARCO and HotpotQA use human passage-level labels, none sentence/claim-level support); **(ii) generated-answer attribution** with *human* support labels (AttributionBench (Li et al., 2024): is a generated answer sentence attributable to its cited evidence?); **(iii) fact-check entailment** with *human* support labels (VitaminC (Schuster et al., 2021): is a short claim supported by its evidence?). For (ii, iii) we score each (claim/sentence, evi-

dence) pair and report AUROC (clustered bootstrap CIs); the clean NLI baseline uses a non-FEVER MNLI model to avoid VitaminC leakage, while the provenance ranker (c) uses an off-the-shelf DeBERTa MNLI/FEVER/ANLI head (so the NLI *column* differs by construct). Table 2 is the headline.

Metric-family performance differs across the audited constructs and datasets. For fact-check entailment we have only one audited boundary dataset, VitaminC ($n=4000$ human labels), where the entailment-trained models score above all no-fine-tuning baselines: FEVER NLI 0.894, clean NLI 0.811, MiniCheck 0.810, versus the surface metrics clustered near 0.60–0.63 (BERTScore 0.596, lexical 0.619, combined 0.628). We treat this as a one-dataset contrast on short edited claims, *not* a general law about fact-checking. For provenance/topicality we report only passage-ranking results (ASQA provenance proxy; human passage-level labels for MS MARCO/HotpotQA), reaching 0.73–0.97 top-1 accuracy (within overlapping bootstrap CIs of NLI, which is however unreliable on ASQA at 0.585); these labels are distinct from sentence/claim-level support and are not pooled with the support tasks. The coarse separation thus holds only loosely and only on these audited benchmarks.

But the “attribution” label is not a reliability boundary. AttributionBench ($n=1610$ human labels) aggregates four source datasets the literature files under one task — is a generated answer sentence attributable to its cited evidence? — and a practitioner who reads “attribution metric” expects a single applicable choice. We make no claim that the four are cognitively identical (they differ sharply in evidence length and structure, which we use below to characterize *where* metrics fail); our point is the practitioner-facing one: this shared label does *not* predict which metric to trust. The best metric is not stable across the four, and the evidence that isolates this *crossover* is rank-based: the per-dataset metric rankings are discordant (Kendall’s $W=0.07$; rank-permutation $p=0.029$), the AttributedQA-vs-LFQA ranking literally inverts (Kendall $\tau=-0.64$, $p=0.031$), and the best-vs-runner-up sign flips (NLI–BERTScore = -0.378 on LFQA vs. $+0.213$ on AttributedQA, CIs exclude 0). A pure dataset-difficulty (main) effect cannot produce a rank reversal. A two-way decomposition of the AUROC matrix places 35% of the cross-cell variance in the interaction

residual (a residual permutation that holds each dataset’s difficulty fixed is borderline, $p=0.053$), while most of the raw cross-cell spread (61%) is dataset-difficulty main effect: some datasets are simply harder for every metric. (An exchangeability permutation that shuffles the dataset assignment of per-example pairs rejects at $p < 0.002$, but that test conflates the main effect with the interaction, so we rely on the rank-based statistics above.) The clean MNLI scorer is the best metric on short-claim AttributedQA (0.904; $+0.270$ over lexical Jaccard, paired 95% CI $[0.198, 0.347]$) yet falls to AUROC 0.531 (chance = 0.50) on long-form LFQA, where BERTScore reaches 0.909 (MNLI–BERTScore = -0.378 , CI $[-0.500, -0.238]$). This collapse pins to one *checkpoint*, not the entailment family: on the same LFQA, FEVER-trained NLI scores 0.720 and the checker MiniCheck 0.852, so “no single metric transfers” is about individual scorers, not a verdict on entailment models as a class. This collapse persists when the NLI token limit increases from 256 to 512 (LFQA 0.531 $\rightarrow$ 0.527). Length controls leave residual dataset differences: at matched evidence length (89–180 words) MNLI still differs by source (0.934 AttributedQA vs. 0.597 ExpertQA), and the within-source short-vs-long drop ($+0.06$ LFQA, $+0.09$ ExpertQA) is far below the 0.38 cross-dataset flip. A purpose-built checker (MiniCheck, 2048-token window) repairs the long-form failure (LFQA 0.852) and has the lowest mean leave-one-dataset-out regret *among the metrics we tested* (AttributionBench mean AUROC 0.774), yet is not uniformly best, losing to BERTScore on Stanford-GenSearch (-0.055 , CI $[-0.104, -0.010]$). Committing to one metric everywhere still incurs modest per-dataset *regret* vs. oracle selection (mean 0.044, CI $[0.014, 0.072]$) — modest in that it is well below the per-dataset sign-flips (0.378) that drive the headline, but its CI excludes zero, so it is a real if small cost. ExpertQA is an underspecified dataset where every metric is weak (oracle best-of-eight only 0.618, barely above chance); three supplementary metrics agree (cross-encoder 0.58, MiniCheck-FT5 0.60, AlignScore 0.60; App. D), so the weakness is not isolated to one scorer, and we lean no claim on it.

An external boundary: long-form alone does not predict the NLI failure. To test whether “long-form” explains the LFQA collapse, we score the full zoo on HAGRID (Kamalloo et al., 2023), an *independent* long-form attribution set (not an Attri-

Construct	Benchmark (label type)	off-the-shelf (no fine-tuning)					trained		
		Lex	MiniLM	MPNet	Comb	BERTSc	NLI _{cl}	NLI _{fe}	MiniChk
(a) <i>Generated-answer attribution</i> — human support labels, AUROC									
AttributedQA (short claims)	AttributionBench (Li et al., 2024)	0.634	0.813	0.766	0.807	0.692	0.904	0.897	0.884
LFQA (long-form)	AttributionBench	0.802	0.739	0.771	0.764	0.909	0.531	0.720	0.852
Stanford-GenSearch	AttributionBench	0.818	0.800	0.779	0.828	0.838	0.822	0.765	0.783
ExpertQA (underspecified)	AttributionBench	0.539	0.608	0.618	0.603	0.569	0.565	0.608	0.577
HAGRID (long-form, indep.)	Kamalloo et al. (2023)	0.752	0.719	0.750	0.722	0.793	0.800	0.787	0.791
(b) <i>Fact-check entailment</i> — human support labels, AUROC									
VitaminC (short claims)	VitaminC (Schuster et al., 2021)	0.619	0.613	0.609	0.628	0.596	0.811	0.894	0.810
(c) <i>Provenance / topicality</i> — passage-level labels, top-1 acc.									
provenance proxy	ASQA (Stelmakh et al., 2022)	0.775	—	0.934	0.919	—	0.585	—	—
human is_selected	MS MARCO (Nguyen et al., 2016)	0.683	0.646	—	0.726	—	0.708	—	—
human support-facts	HotpotQA (Yang et al., 2018)	0.959	0.938	—	0.972	—	0.979	—	—

Table 2: **Benchmark-by-benchmark audit; constructs are reported separately and never pooled.** The main multi-dataset non-portability result is the four-source AttributionBench audit in (a); HAGRID, VitaminC, and the provenance benchmarks are independent boundary/contrast checks, not evidence for a universal metric ranking. AUROC for the support tasks (a, b; binary support against *human* sentence/claim labels) and top-1 *passage-ranking* accuracy (not AUROC) for provenance (c; passage-level labels — a provenance proxy for ASQA, human passage-relevance for MS MARCO/HotpotQA — which are *not* sentence/claim-level support judgments); best per row in bold. For the human-support tasks (a, b) NLI_{cl} is the *clean, non-FEVER* MNLi model (roberta-large-mnli) and NLI_{fe} is FEVER-trained; MiniChk is MiniCheck (Tang et al., 2024). For the human-support tasks (a, b), Comb is the two-term blend $0.47 \cdot \text{lex} + 0.53 \cdot \text{sem}$ over each (claim/sentence, evidence) pair (the answer-hint and citation terms of Eq. (1) do not apply — support pairs carry no gold answer or inline citation — and are omitted); MiniLM is the semantic encoder there. The provenance NLI column (c) is the off-the-shelf DeBERTa-v3 MNLi/FEVER/ANLI head used as a ranker (Section 3.1). **Coarsely**, metric-family performance differs across the audited settings: entailment-trained models score above all no-fine-tuning baselines on fact-check (b), while off-the-shelf relevance is within overlapping CIs of NLI on provenance (c). **But within one construct** (a), the best metric is not stable across the four AttributionBench source datasets (the per-dataset metric rankings are discordant: Kendall $W=0.07$, $p=0.029$; the AttributedQA-vs-LFQA ranking inverts, $\tau = -0.64$; HAGRID is an independent long-form set): the clean MNLi scorer is best on short-claim AttributedQA (0.904) yet AUROC 0.531 (chance = 0.50) on long-form LFQA, where BERTScore wins (0.909). The LFQA collapse persists at both tested NLI token limits (256, 512). Length-controlled analyses leave residual dataset differences but do not isolate their cause. MiniCheck has the lowest mean leave-one-dataset-out regret *among those tested* (AttributionBench mean AUROC 0.774; regret 0.044) but still not uniformly best. ExpertQA is an underspecified dataset (oracle best-of-eight only 0.618). The semantic provenance ranker (c) is all-mpnet-base-v2 for the ASQA confusable test (Section 3.1; hence its 0.934 sits in the MPNet column) and all-MiniLM-L6-v2 for MS MARCO/HotpotQA; BERTScore, NLI_{fe}, and MiniCheck are AUROC support scorers and are not run as provenance *rankers* (c). Human-labeled *support* (a, b) and proxy-labeled *provenance* (c) are distinct constructs and are not pooled.

butionBench source) with human sentence/claim-level attributable labels ($n=2150$). Here the clean MNLi scorer is instead the *best* metric (0.800; BERTScore 0.793, MiniCheck 0.791, lexical 0.752). HAGRID answers have LFQA-comparable claim length but much shorter, explicitly [n]-cited evidence (short cited quotes rather than LFQA’s ~ 320 -word passages). This refutes the simplest reading: that NLI fails whenever the *answer* is long-form. It does *not*, however, rule out long *evidence* (which HAGRID lacks) as the mediator; if anything, HAGRID’s short cited evidence is consistent with our evidence-length account below. Rankings are thus dataset-specific even across superficially similar long-form settings; we cannot isolate the causal latent characteristic from dataset

identity here, and leave that to future work.

We also include BEGIN (Dziri et al., 2022) only as an appendix boundary check in a related but different setting (knowledge-grounded *dialogue* with response-level labels; App. E). Its pattern is consistent with dataset dependence, but it is not part of the main multi-dataset audit, not used in the transfer criterion, and not co-equal with the AttributionBench/HAGRID evidence. We caution that the provenance results (construct i) use passage-level provenance/relevance labels, not human sentence/claim-level support judgments, and we do not pool them with (ii, iii).

Naive cross-dataset metric selection does not generalize. Can one *learn* a good default from observed datasets? Picking the metric with the

best mean AUROC on seen datasets fails leave-one-dataset-out over the four sources: mean realized regret is 0.172 AUROC, and the learned rule is on average 0.060 AUROC *worse* than fixing the raw MNLI metric. The loss is *not* on the LFQA fold (there the rule picks MNLI and ties it); it comes from the other folds — most sharply AttributedQA-held-out, where the rule picks BERTScore (0.69) over MNLI (0.90), a -0.21 swing — so averaging over seen datasets actively mis-selects on the held-out one. The actionable consequence is a *validation discipline*: specify the target construct; compare candidate scorers on a representative sample with target-dataset human labels; then select and report the scorer with its uncertainty and exclusions. Dataset properties alone do not establish a selection rule here. These conclusions are bounded to the four datasets and eight metrics audited here. A minimax rule minimizes the largest regret over the *seen* datasets. Using all four datasets, it selects MiniCheck, as does best-on-average (worst-case regret 0.058). Refitting on the three seen datasets in each leave-one-dataset-out fold gives the same choices as best-on-average and the same 0.172 mean held-out regret. This minimax rule does not improve held-out performance; other robust selectors and ensembles are not evaluated.

When can LLM adjudication stand in for human labels? Since target-dataset human labels are often missing, we probe cross-family LLM adjudication: two frontier models from *different* families (Claude Opus 4.8, GPT-5.4) independently annotate 160 AttributionBench items (40/source), reported as *LLM-adjudicated*, never human/gold. The families agree with *each other* ($\kappa=0.832$) but only moderately with *human* labels ($\kappa\approx0.47$), and that human-agreement is itself dataset-dependent (accuracy AttributedQA 0.93/0.88 down to ExpertQA 0.58/0.63); on Stanford they agree strongly with each other ($\kappa=0.90$) yet diverge from humans, so high inter-model agreement is *not* evidence of validity. LLM adjudication is thus a corroborative proxy only in the easiest (short-factoid) dataset.

Is a prompt-based LLM judge more stable across datasets? Our non-transfer finding is about *automatic* scorers (including trained ones: clean/FEVER NLI, MiniCheck). The natural rejoinder — practitioners increasingly use a prompt-based *LLM judge* — we test directly (Opus 4.8 as a scoring metric, Table 14): on the same four Attri-

butionBench sources it never drops to the chance-level collapses the strong automatic scorers suffer (AUROC 0.73–0.92, vs. clean MNLI 0.53–0.90 and BERTScore 0.57–0.91 on those four) and does *not* collapse on LFQA. The LLM judge *avoids the collapses* here, but this is a boundary observation, not an escape hatch: it is still not uniformly best, costs $\sim 100\times$ more, and is non-deterministic; its AUROC is computed over 408 of 480 sampled items with valid parsed scores (15% unscored). Assigning 0.50 to unscored items retains above-chance AUROC point estimates in all four datasets (Table 14), but does not rule out selection bias. The judge relocates rather than removes the validation burden.

We now detail the provenance benchmarks behind Table 2(c), whose labels are passage-level (a provenance proxy for ASQA, *human* passage-relevance for MS MARCO and HotpotQA), *not* sentence/claim-level support. On **MS MARCO** v2.1 (Nguyen et al., 2016) (11,836 dev queries with ≥ 3 passages, a well-formed answer, and a human-selected passage; “combined” is the gold-free $0.47\text{ lex} + 0.53\text{ sem}$ blend), we rank each query’s own passages and ask whether the top-1 is *human*-selected (*is_selected*). Over 2000 queries/seed $\times 3$, all scorers far exceed the random baseline (0.104): lexical 0.683, semantic 0.646, NLI 0.708, combined 0.726 (combined and NLI within bootstrap noise, both beating the components), so the score tracks *human* passage relevance, a construct-(i) sanity check, not grounded support. Under the confusable protocol ($n=300\times 4$) lexical-only top-1 is 0.885, semantic 0.995, combined 0.996 (combined–lexical $+0.111$, every per-seed CI excludes zero), and the added NLI baseline reaches 0.952 — above lexical, below semantic. The same lexical-fooled, semantic-survives ordering as ASQA, near-ceiling because MS MARCO is more extractive.

On **HotpotQA** (Yang et al., 2018) (distractor setting), with *human* supporting-fact annotations over ten candidate paragraphs (1500/seed $\times 3$), all scorers far exceed the random baseline 0.201 (Table 9c); here, on short multi-hop answers reusing the question’s entities, *lexical* and *NLI* overtake *semantic*: the mirror image of ASQA.

Within provenance, no single component is universal; the blend is a convenient baseline. The three *provenance* benchmarks complicate any sim-

ple “semantic wins” summary: each single component has a dataset where it is weakest: semantic similarity on short, entity-heavy multi-hop answers (HotpotQA), and lexical overlap and off-the-shelf NLI on synthesized long-form answers (ASQA, where NLI falls to 0.585). The *combined* score stays at or near the top across these *provenance* tasks (Tables 2c, 9; on the human *is_selected* gate it leads at 0.726, and on the MS MARCO confusable top-1 test it reaches 0.996): within construct (i) only, the fixed blend is a convenient provenance baseline that stays at or near the top across these retrieval datasets. This blend is *not* a general-purpose *grounding* metric: on the human-support tasks (constructs ii, iii) it is far from best (Table 2), consistent with our headline of non-portability in the core generated-answer attribution audit. The provenance behavior is stable across encoders and distractor counts: all-MiniLM-L6-v2 gives ASQA semantic accuracy 0.919–0.939 across $k \in \{3, 8, 10\}$ (vs. all-mpnet-base-v2’s 0.934 at $k=5$), with the combined–lexical gap positive (+0.13 to +0.15) throughout.

5 Conclusion

Under our operational near-best screen, no audited *automatic* scorer is consistently near-best across the datasets of generated-answer attribution, the construct with the most multi-dataset human-labeled coverage (the four AttributionBench sources, $n=1610$, with independent HAGRID, $n=2150$). Metric-family performance varies across the audited settings, but within answer attribution the best metric is unstable across datasets (per-dataset rankings discordant, Kendall $W=0.07$, $p=0.029$; the NLI scorer flips from best on AttributedQA to chance on LFQA), and even the lowest-regret metric leaves measurable per-dataset regret. A prompt-based LLM judge avoids the chance-level collapses (AUROC 0.73–0.92, no LFQA collapse) but is costlier and non-deterministic, relocating rather than removing the problem. Evaluations should report their dataset and construct and validate the metric on them.

6 Limitations

Construct and domain shift are entangled. Our largest-gap “NLI wins” contrast (0.81–0.89 vs. ≈ 0.60 AUROC) is also our only out-of-domain benchmark (VitaminC), so we cannot fully sepa-

rate construct from domain shift. The in-domain AttributionBench result (where the gap closes) supports dependence on the evaluated setting, not an isolated construct effect or a claim that entailment is intrinsically superior. **Three sentence/claim-level human-support benchmarks, all English.** Only AttributionBench, HAGRID, and VitaminC carry human *sentence/claim-level* support labels (BEGIN, App. E, has human labels but at the *response* level — an external boundary, not a main construct); all are English. Broader in-construct human labels (other domains/languages) are the natural next step. **Provenance uses passage-ranking labels, not the sentence-attribution task of (ii, iii).** We pose construct (i) as passage ranking: ASQA uses a provenance proxy, while MS MARCO (*is_selected*) and HotpotQA (supporting-fact annotations) provide human relevance labels. We use all three only to ask whether a score ranks a relevant passage top, *not* to predict sentence/claim-level support judgments. We keep these strictly separate from the human-support constructs and never pool them. **Correctness confound.** The full-dev provenance gap is partly correctness-confounded (mitigated, not removed, by the matched subset). **Item-level, not system-level.** Our consequence is at the level of choosing an evaluator under dataset uncertainty (per-item AUROC and the failure of a naive cross-dataset selector); we do *not* show that this metric instability reverses *system* rankings across generators. Whether item-level non-transfer induces model-selection reversals is important future work.

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A ASQA Internal Consistency Checks

For completeness we report two ASQA checks that we treat as internal consistency checks, *not* validation, because their labels are threshold heuristics

Gate	Metric	Result	Note
1	top-1 retrieval (comb.)	0.966	lex 0.915 / sem 0.966
2	matched-pair agreement	10/10	gap +0.185
3	full-dev gap ($n=623$)	+0.164	CI[+0.152,+0.177]
3	matched subset ($n=98$)	+0.160	CI[+0.133,+0.189]
3	evidence-prec. gap	+0.234	—

Table 3: Three-gate internal-consistency check on ASQA dev. Labels are heuristic/proxy, not human. The correctness-matched subset (Gate 3, row 4) shows the largest gap.

Gate-3 metric	Canonical ($w_h=0.10$)	Hint = 0 (ablated)
Matched-subset gap	+0.160	+0.153
95% CI	[+0.133, +0.189]	[+0.127, +0.180]
Full-dev gap	+0.164	+0.146
95% CI	[+0.152, +0.177]	[+0.134, +0.157]
Evidence-prec. gap	+0.234	+0.205

Table 4: Answer-hint ablation on Gate 3 (ASQA full dev, $n=623$; matched subset $n=98$). Removing the label-aware answer-hint term ($w_h=0$, other weights fixed) leaves the grounding gap essentially intact, with the matched-subset CI still excluding zero. The gap is not an artifact of the label-aware term.

derived from the same lexical/semantic features under test (hence circular). **(1) Gold-slice retrieval:** on a 20-example slice with 2–3 distractor passages, top-1 retrieval is 0.966 combined / 0.915 lexical / 0.966 semantic against heuristic “gold” labels. **(2) Matched-pair agreement:** on 10 strong/decent pairs the score’s preference agrees 10/10 with a lexical-overlap groundedness proxy (gap +0.185). Both are corroborative only; the human-labeled constructs in Table 2 carry the paper’s claims. The full gold-reference vs. unadapted-model gates and the answer-hint ablation are in Tables 3–4.

B Per-Component Ablation

Table 5 reports a leave-one-out ablation of the four score components on the Gate-3 strong—decent gap. It complements the lexical-weight sweep (Section 3.1) and the answer-hint ablation (Section 3.1): the lexical and semantic terms each carry a substantial share of the gap, while the answer-hint and citation terms are minor, consistent with the main-text finding that the score’s signal does not depend on the label-aware hint.

Weighting	Matched gap	Full-dev gap
Canonical (.40/.45/.10/.05)	+0.160	+0.164
drop lexical	+0.062	+0.075
drop semantic	+0.097	+0.101
drop hint	+0.153	+0.146
drop citation	+0.168	+0.170
Lexical-only	+0.247	+0.222
Semantic-only	+0.139	+0.138

Table 5: Per-component leave-one-out on the Gate-3 strong—decent attribution gap (ASQA full dev; matched subset $n=98$, full dev $n=623$; all matched CIs exclude 0). Dropping lexical or semantic shrinks the gap most (both contribute); dropping the label-aware hint barely changes it ($+0.160 \rightarrow +0.153$, confirming the gap is not a hint artifact); dropping citation is harmless. The lexical-only and semantic-only rows show each of those two components carries a gap on its own.

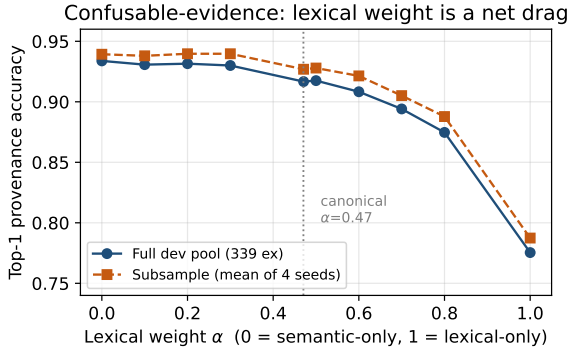


Figure 1: Lexical-weight ablation under confusable evidence (construct i). Top-1 provenance accuracy is flat at low α and then declines as the lexical weight α rises; the canonical blend ($\alpha \approx 0.47$) is already past the knee. Semantic-only ($\alpha=0$) is best.

C A Grounding-Reward Stress Test: Do Answer-Level Metrics Detect a Large Rewrite?

To test whether answer-level metrics detect a large grounding-targeted rewrite, we ran a stress test. We fine-tuned the generator (GRPO (Shao et al., 2024) on Qwen2.5-{3B, 7B}-Instruct) with a reward that directly targets evidence grounding: it scores each answer sentence by whether its claims are supported by the provided evidence and whether that support *counterfactually* depends on the evidence being present (rather than on the model’s priors), along with evidence coverage and epistemic commitment. Comparing the adapted model against the *unadapted* base (a base-vs-base+LoRA comparison, not SFT-vs-RL) on all 948 ASQA dev answers, at 3B the reward rewrites 89.9% of generations, yet QA-EM, ROUGE-L, and citation count stay flat.

This is the paper’s measurement point in miniature: a large, grounding-directed intervention on the output leaves the answer-level metrics we test here flat, so those metrics are insensitive to the kind of grounding-targeted rewrite this intervention produces. We use the reward only as a probe, not as evidence that true grounding improved.

The metrics do not move. At 3B, QA-EM (corrected ASQA str_em, any-alias match) is 0.299 for the base and 0.297 for the adapted model (delta -0.002 , 95% paired-bootstrap CI $[-0.011, +0.008]$): no statistically meaningful change. ROUGE-L moves $+0.006$ and citation count is flat. At 7B the pattern repeats (QA-EM $0.377 \rightarrow 0.381$, CI $[-0.004, +0.012]$). Under a legacy substring metric the 7B values are $0.240 \rightarrow 0.241$, the same flat pattern. We measure citation *count*, which is flat; we did not regenerate NLI-based ALCE citation recall/precision (Gao et al., 2023) (it requires a large entailment model and a different decoding setup), and we make no claim about it here.

The intervention is not vacuous, and its scope is narrow. At 3B the reward alters 89.9% of the 948 generations (mean token Jaccard 0.59; 81.2% differ substantially, Jaccard < 0.9), yet QA-EM, ROUGE-L, and citation count register essentially nothing. We do *not* claim the reward improved grounding (a single untuned run could be weak); it shows only that these answer-level metrics are insensitive to a large grounding-targeted rewrite. The scope is therefore narrow: the probe does not establish that these metrics would miss a *genuine* grounding improvement; it is motivation, not proof, and the audit does not depend on it.

Setup. Table 6 lists the LoRA and GRPO configuration, recovered from the saved adapter configs and the training setup; training-time loss/reward trajectories were not retained. Table 7 reports the metric deltas. The regeneration uses greedy decoding: the flatness reproduces, though absolute QA-EM differs from prior temperature-sampled prose.

D Supplementary Metric Families

Beyond the eight metrics of Table 2, we ran three additional supplementary metrics on the AttributionBench sources to check that the per-source pattern is not specific to our metric set: a cross-encoder reranker

Setting	Value
<i>Base models</i>	
Backbone	Qwen2.5-{3B, 7B}-Instruct
Adapter	LoRA (PEFT), CAUSAL_LM
<i>LoRA</i>	
Rank r	8
α	16
Dropout	0.0
Target modules	q, k, v, o, gate, up, down_proj
<i>GRPO (TRL GRPOTrainer)</i>	
Configuration	DAPO-style
Generations / prompt	4
Max completion length	512 tokens
Input truncation	2048 tokens
Reward weights	fixed (not tuned)
<i>Held-out evaluation</i>	
Decoding	greedy
Eval set	full ASQA dev (948)

Table 6: Grounding-reward training and evaluation configuration, recovered from the saved LoRA adapter configs and the GRPO setup. We report these for methodological transparency; the reward is a probe (App. C), not a proposed method. Training-time loss/reward trajectories were not retained with the saved adapters and are not reported.

Model	QA-EM	ROUGE-L	Cites
<i>3B</i>			
base (unadapted)	0.299	0.219	1.05
+ grounding reward	0.297	0.225	1.04
Δ QA-EM -0.002, CI[-0.011,+0.008]			
<i>7B</i>			
base (unadapted)	0.377	0.191	1.83
+ grounding reward	0.381	0.193	1.83
Δ QA-EM +0.004, CI[-0.004,+0.012]			

Table 7: Grounding-reward stress test on full ASQA dev (948 ex, greedy decoding). QA-EM is corrected ASQA str_em. The reward produces no statistically meaningful change at either scale.

(ms-marco-MiniLM-L6), MiniCheck-FT5 (flan-t5-large), and AlignScore-large (Zha et al., 2023), a metric purpose-built for factual consistency. Table 8 shows all three reproduce the pattern, and in particular all are near chance on ExpertQA (AUROC 0.58–0.60, chance = 0.50), so the weak performance there is not isolated to the originally audited scorers.

E BEGIN: Appendix Boundary Check (Dialogue, Response-Level Labels)

As an external, different-domain check (knowledge-grounded *dialogue* rather than QA), we score the metric zoo on BEGIN (Dziri et al., 2022),

Dataset	Cross-enc.	MiniChk-FT5	AlignScore
AttributedQA	0.867	0.896	0.901
LFQA	0.853	0.859	0.811
Stanford-GenSearch	0.836	0.795	0.784
ExpertQA	0.580	0.603	0.596

Table 8: **Three supplementary metrics reproduce the per-source pattern** (AUROC on AttributionBench human attribution labels): a cross-encoder reranker (ms-marco-MiniLM-L6), MiniCheck-FT5 (flan-t5-large), and AlignScore-large (Zha et al., 2023). All three reach AUROC 0.867–0.901 on AttributedQA, weaker but usable on LFQA/Stanford, and weak on ExpertQA (AUROC 0.58–0.60) — matching the eight metrics of Table 2. ExpertQA is weak for every scorer we tried (AUROC 0.58–0.62, only just above chance), so its low ceiling is not isolated to any single metric.

$n=11,466$ system responses with human “fully attributable” vs “not” labels (we drop the “generic” class), using knowledge as evidence and the response as the claim. Table 10 shows the grounding-trained family (clean MNLI, MiniCheck) has top AUROC on every dataset here — the opposite of LFQA, where clean MNLI collapsed. This is a boundary condition that sharpens, rather than contradicts, non-portability: an otherwise top-scoring family still has a dataset (LFQA) where it fails, so no single metric is the lowest-regret fixed choice across datasets. We treat BEGIN’s response-level binary labels as related to, but distinct from, the sentence/claim-level attribution of the main constructs, and do not pool them.

F Independent Replication and Robustness

Independent multi-dataset replication. To test whether non-transfer holds beyond AttributionBench, we score the same zoo on eight *disjoint* LLM-AggrFact source datasets (excluding the two, ExpertQA and LFQA, that overlap our audit; $n=7336$ pairs). No metric transfers there either (Table 11): four different metrics win across the eight datasets and the same clean MNLI scorer swings AUROC 0.494 $\rightarrow$ 0.946 — the winning metric changes by dataset, the signature of a genuine crossover rather than a uniform difficulty shift.

Portability-criterion sensitivity. The non-transfer conclusion does not hinge on the CI-relative screen: an ϵ -regret sweep, Kendall- τ rank-inversion, a cluster-respecting permuta-

(a) Human passage-relevance gate (top-1 is_selected)		
Scorer	Top-1 acc.	vs. random
Random passage	0.104	—
Lexical-only	0.683	+0.579
Semantic-only	0.646	+0.542
NLI entailment	0.708	+0.604
Combined (canon.)	0.726	+0.622

(b) MS MARCO confusable test (top-1 provenance)		
Scorer	Top-1 acc.	Fooled
Lexical-only	0.885	0.115
Semantic-only	0.995	0.005
Combined (canon.)	0.996	0.004
NLI entailment [†]	0.952	0.048

(c) HotpotQA human supporting-fact gate		
Scorer	Top-1 acc.	vs. random
Random paragraph	0.201	—
Lexical-only	0.959	+0.758
Semantic-only	0.938	+0.737
NLI entailment	0.979	+0.778
Combined (canon.)	0.972	+0.771

Table 9: Additional benchmarks (best in bold). (a) MS MARCO human passage-relevance gate: rank a query’s retrieved passages per answer sentence; ground truth is the *human is_selected* label (passage-level, not sentence/claim-level support), 2000 queries $\times$ 3 seeds. (b) MS MARCO confusable-evidence test (Section 3.1 design), $n=300 \times 4$ seeds: the combined/semantic advantage over lexical holds (gap +0.111; four-seed gaps +0.097 to +0.130, all CIs >0). (c) HotpotQA distractor-setting human supporting-fact gate, 1500×3 seeds: rank a human-supporting paragraph top (2-hop, so ≈ 2 supporting among 10; random 0.201). On these extractive/multi-hop answers lexical and NLI overtake semantic — the mirror image of ASQA — yet the combined score stays near-best on every benchmark. [†]NLI is a separate $n=200 \times 3$ subsample.

tion, paired best-vs-rest tests, and threshold-free AUPRC/balanced-accuracy all agree (Table 12).

A descriptive correlate of the collapse. Evidence length / sentence count tracks where clean MNLI fails (Table 13), consistent with single-premise entailment degrading under cross-sentence aggregation (Laban et al., 2022); we report this as descriptive, not causal.

Is a prompt-based LLM judge more stable across datasets? We also run a prompt-based LLM judge (Opus 4.8) as a scoring metric across the four AttributionBench sources (Table 14). On those same four sources it never drops to the chance-level collapses the strong automatic scorers suffer (range 0.731–0.918 vs. clean MNLI’s 0.531–0.904 on the four) and does not collapse on LFQA, but is not uniformly best, costs $\sim 100\times$ more, and is non-deterministic. Its complete-case AUROC uses

Metric	cmu-dog	TopicalChat	Wow	pooled
lexical	0.743	0.668	0.846	0.766
sem(MiniLM)	0.747	0.565	0.826	0.756
sem(MPNet)	0.772	0.616	0.870	0.780
combined	0.760	0.640	0.860	0.764
BERTScore	0.762	0.833	0.845	0.799
NLI _{fe}	0.806	0.687	0.896	0.799
NLI _{cl}	0.880	0.756	0.944	0.890
MiniCheck	0.922	0.877	0.885	0.883

Table 10: **BEGIN dialogue-attribution** (Dziri et al., 2022) AUROC (knowledge-grounded dialogue; $n=11,466, 2,878$ attributable; best per column in bold). In this *new domain*, the grounding-trained family (NLI_{cl}, MiniCheck) is uniformly strongest, in contrast to the QA generated-answer attribution of Table 2(a) where NLI_{cl}’s rank was unstable. BEGIN is a boundary condition — a dataset where these metrics rank at the top across all three sub-domains — which is consistent with (not a contradiction of) non-portability: the same NLI_{cl} collapses on LFQA.

408 of 480 sampled items with valid parsed scores. The other 72 (15%) are unscored; the saved scores do not distinguish refusals, truncation, parsing failures, and API errors. Non-random exclusions could inflate performance and its apparent stability advantage. Table 14 reports counts and a sensitivity check assigning 0.50 to every unscored item: all four AUROC point estimates remain above chance. This check does not resolve selection bias or establish superiority over automatic scorers, which were evaluated on the full source datasets rather than the judge’s retained subsets. The judge remains subject to target-dataset validation.

LLM-judge prompt templates. For reproducibility we give the exact prompts verbatim (Figure 2). The judge runs as a *scoring metric*: claude-opus-4-8, API-default decoding (no temperature parameter supplied), maxTokens= 20, evidence/claim/question truncated to 1500/600/300 characters; the support probability is parsed with a regex and clamped to $[0, 1]$. Unscored items are excluded from the primary AUROC and imputed only for the sensitivity check. The separate cross-family *annotation* probe (Section 4; Opus 4.8 and GPT-5.4, $n=160$) instead elicits a discrete label, scored as inter-annotator agreement, never as gold.

Cross-benchmark landscape. Pooling per-dataset AUROCs across all 17 human-labeled datasets we audit (Table 15), six different metrics are best on at least one dataset, the most frequent winner (MiniCheck) is best on only 6/17 (35%), and every metric’s AUROC swings by 0.34–0.45

(a) Graded scoring prompt (LLM judge as a metric):

You are an automated ATTRIBUTION-scoring rubric for a QA system. Given a CLAIM (a sentence from a generated answer) and the EVIDENCE it cites, rate how fully the claim is supported by (attributable to) the evidence.

Output EXACTLY one line and nothing else, even if uncertain:
SUPPORT: <number 0.00-1.00>

1.00 = every factual element of the claim is directly supported by the evidence; 0.00 = the evidence is unrelated to or contradicts the claim; intermediate values for partial support. Use the full range and be calibrated. This is an automated rubric: always return a number (use 0.50 if truly unsure). Do not refuse and do not explain.

QUESTION: {question}
CLAIM: {claim}
EVIDENCE: {evidence}

(b) Discrete annotation prompt (cross-family κ probe):

You are judging ATTRIBUTION for a question-answering system. Given a CLAIM (a sentence from a generated answer) and the EVIDENCE it cites, decide: is the claim fully supported by (attributable to) the evidence?

Answer STRICTLY in this format on one line:
LABEL: <attributable|not_attributable>
CONF: <0-1>

A claim is "attributable" only if all of its factual content is supported by the evidence. If the evidence is unrelated, contradicts, or only partially supports the claim, answer not_attributable.

QUESTION: {question}
CLAIM: {claim}
EVIDENCE: {evidence}

Figure 2: **Verbatim LLM prompt templates.** (a) The graded support-probability prompt used to run Claude Opus 4.8 as a scoring metric (Table 14); (b) the discrete-label prompt used by the $n=160$ cross-family adjudication probe (Opus 4.8 and GPT-5.4). {question}, {claim}, and {evidence} are filled per item. Both are disclosed as LLM outputs, never human/gold.

across datasets. (Descriptive landscape; constructs are tagged and not pooled into a single test.)

Dataset	Lex	Comb	BERTSc	NLI _{cl}	NLI _{fe}	MiniChk
AggreFact-CNN	0.723	0.730	0.771	0.584	0.611	0.672
AggreFact-XSum	0.500	0.548	0.543	0.625	0.714	0.784
ClaimVerify	0.603	0.636	0.693	0.605	0.655	0.764
FactCheck-GPT	0.605	0.742	0.727	0.819	0.828	0.833
Reveal	0.597	0.813	0.661	0.946	0.923	0.944
TofuEval-MediaS	0.605	0.641	0.695	0.675	0.635	0.686
TofuEval-MeetB	0.659	0.685	0.642	0.552	0.610	0.674
Wice	0.538	0.504	0.549	0.494	0.553	0.708

Table 11: **Independent replication: no audited metric transfers across a second, disjoint multi-dataset benchmark.** AUROC against *human* support labels on eight LLM-AggreFact (Tang et al., 2024) source datasets ($n=7336$ pairs), excluding the two (ExpertQA, LFQA) that overlap our AttributionBench audit so the panel is independent; best per row in bold. As in Table 2, no single audited scorer is best across datasets: four different metrics win the eight rows — BERTScore (2), MiniCheck (4), clean MNLI (Reveal), and the lexical/semantic blend (Comb, TofuEval-MeetB) — the same clean MNLI scorer swings from AUROC 0.494 (Wice, $\leq$ chance) to 0.946 (Reveal), so the winning metric changes by dataset, and committing to the lowest-regret single metric (MiniCheck) still leaves mean leave-one-dataset-out regret 0.015 (max 0.099). This is a faithfulness/grounding benchmark (claim-vs-document support), adjacent to but distinct from sentence-level citation attribution; we report it as an independent *boundary replication* and do not pool it with the AttributionBench audit.

<i>(a) Portability-criterion sensitivity (AttrBench 4 sources)</i>	
ϵ -regret tolerance	# metrics “transfer”
$\epsilon \in \{0, .01, .02, .05\}$	0
$\epsilon = 0.10$	1 (MiniCheck)
<i>(b) Cross-dataset rank stability (Kendall τ)</i>	
mean τ over dataset pairs	−0.20
AttributedQA vs. LFQA	−0.64 ($p=.031$)
<i>(c) Interaction, main-effect-controlled</i>	
rank concordance (Kendall W)	0.07, $p=.029$
residual perm. (difficulty fixed)	$p=.053$
<i>mis-spec. shuffle, ref. only</i>	$p<.002$
<i>(d) Paired best-vs-rest on LFQA (BERTScore best)</i>	
BERTScore – clean MNLI	+ .378 [.24, .50], $p<.001$
BERTScore – MiniCheck	+ .058 [−.02, .14], $p=.127$

Table 12: **The non-transfer conclusion does not hinge on the CI-relative criterion.** Four independent readings agree on AttributionBench’s four sources: (a) under an ϵ -regret screen no audited metric is within ϵ of the per-dataset oracle on *every* dataset for any $\epsilon \leq 0.05$; (b) metric rankings are uncorrelated-to-negative across datasets (the AttributedQA/LFQA ranking literally inverts); (c) the crossover survives controlling for dataset difficulty — within-dataset rank concordance is near zero (Kendall $W=0.07$, $p=.029$) and a residual permutation that holds each dataset’s difficulty fixed puts 35% of the cross-cell variance in the interaction residual (borderline, $p=.053$; the looser exchangeability shuffle that conflates difficulty with interaction gives $p<.002$ and is shown for reference only); (d) the headline LFQA reversal is decisive (BERTScore beats clean MNLI by 0.378). **Honestly**, “no single metric transfers” is robust, but the best metric is not always *uniquely* best: on LFQA, BERTScore vs. MiniCheck is not significant ($p = 0.127$). All quantities are recomputed and match the Table 2 AUROCs.

Dataset	words	sent.	NLI _{cl}	BERTSc	MiniChk
AttributedQA	103	4.2	0.904	0.692	0.884
Stanford-GenSearch	82	4.1	0.822	0.838	0.783
ExpertQA	174	7.9	0.565	0.569	0.577
LFQA	321	13.9	0.531	0.909	0.852

Table 13: **A descriptive correlate of the clean-MNLI collapse: evidence length / sentence count.** Mean evidence length (words) and evidence sentence count per AttributionBench source, against per-dataset AUROC. The clean MNLI scorer is strongest where evidence is short and few-sentence (AttributedQA: 103 words, 4.2 sentences $\rightarrow$ 0.904) and collapses where evidence is long and many-sentence (LFQA: 321 words, 13.9 sentences $\rightarrow$ 0.531), while token-level BERTScore moves the opposite way (0.692 $\rightarrow$ 0.909). This is consistent with a single-premise entailment head degrading when support must be aggregated across many sentences (Laban et al., 2022). We report this as a **descriptive** correlate, not a causal mechanism: with four datasets the dataset-level association is underpowered, the per-example abtractiveness signal does not cleanly separate NLI from BERTScore, and ExpertQA is a near-chance exception. Identifying the causal driver is left to future work.

Dataset	Valid n	Judge	Imputed	Best auto.
AttributedQA	101/120	0.918	0.892	0.904
LFQA	105/120	0.878	0.859	0.909
Stanford-GenSearch	100/120	0.794	0.763	0.838
ExpertQA	102/120	0.731	0.708	0.618

Table 14: **LLM-judge AUROC and unscored-item sensitivity.** Claude Opus 4.8 emits a graded support score evaluated against *human* AttributionBench labels. Sampling used 60 positive and 60 negative items per source; 408/480 yielded valid scores. Judge: complete-case AUROC. Imputed: AUROC on all 120/source after assigning 0.50 to unscored items. Exclusions are 19/15/20/18 in row order. Above-chance point estimates persist under this imputation, which does not rule out selection bias. Best auto.: best of the eight automatic scorers on each *full* source dataset (Table 2), not a paired comparison on the judge subset: clean MNLI, BERTScore, BERTScore, and MPNet in row order. The judge is not uniformly best by these point estimates and remains costlier ($\sim 100\times$) and non-deterministic. See App. F for limits.

Metric	min	max	range
lexical Jaccard	0.500	0.839	0.339
MiniLM cosine	0.499	0.841	0.342
MPNet cosine	0.477	0.851	0.374
combined blend	0.504	0.845	0.341
BERTScore	0.543	0.909	0.366
NLI _{fe} (FEVER)	0.553	0.923	0.370
NLI _{cl} (clean MNLI)	0.494	0.946	0.452
MiniCheck	0.577	0.944	0.367

Table 15: **Cross-benchmark landscape: every audited metric swings widely across datasets, and no metric dominates.** Per-metric AUROC range across the **17 human-labeled datasets** we audit, spanning four evaluation settings (5 generated-answer attribution incl. HAGRID; 1 fact-check; 3 dialogue-attribution sub-domains; 8 LLM-AggreFact faithfulness datasets). Six different metrics are best on at least one dataset; the most frequent winner (MiniCheck) is best on only 6/17 (35%). The same clean MNLI scorer spans AUROC 0.494 (below chance) to 0.946 — a 0.45 swing. The best fixed default (MiniCheck, mean AUROC 0.791) still leaves mean leave-one-dataset-out regret 0.026 (max 0.099). This is a *descriptive* landscape pooling the per-dataset AUROCs reported in the paper; datasets span distinct constructs and are *not* pooled into a single significance test.